

The Logic of Inference

150 minutes · 6 self-check questions

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What a p-value is and is not, what a confidence interval actually tells you, and why an effect size belongs in every result you report.

Learning objectives

- Explain sampling variation and the sampling distribution
 - Interpret a confidence interval correctly, and reject the common misreadings
 - State precisely what a p-value measures
 - Distinguish statistical significance from practical importance
 - Calculate and interpret common effect sizes
 - Explain Type I and Type II error, power, and the cost of multiple testing
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Why samples vary

- Any sample is one of many that could have been drawn; each gives a slightly different estimate
 - The sampling distribution is the distribution of an estimate across all possible samples
 - Its standard deviation is the standard error of the estimate
 - The central limit theorem says that for a large enough sample the sampling distribution of the mean is approximately normal, whatever the shape of the underlying data
 - This is why means can be handled with normal-theory methods even when the raw data are skewed
 - It says nothing about the data being normal — a persistent confusion
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Confidence intervals

- A 95% confidence interval is constructed so that, over repeated sampling, 95% of such intervals contain the true value
- It is not "a 95% probability the true value lies in this interval" — the true value is fixed, the interval is what varies
- Its width reflects precision: wide means the study could not pin the effect down
- A CI that includes the null value corresponds to $p > 0.05$, but carries far more information
- Report intervals for every effect estimate, not just p-values

- A non-significant result with a narrow interval around zero is evidence of no meaningful effect; with a wide interval it is simply uninformative

IN PRACTICE

"No significant difference" and "no difference" are not the same claim. The confidence interval is what tells them apart, which is the single strongest argument for always reporting it.

What a p-value is

- The probability of observing data at least as extreme as yours, if the null hypothesis were true
- It is not the probability the null hypothesis is true
- It is not the probability your finding is a fluke
- It is not a measure of effect size — a tiny, unimportant difference gives a small p in a large enough sample
- 0.05 is a convention, not a law of nature, and nothing decisive happens between 0.049 and 0.051
- Report exact values ($p = 0.03$), not thresholds ($p < 0.05$), and never " $p = 0.000$ " — write $p < 0.001$

Effect size

- The effect size answers "how big", which is what a reader actually wants to know
- Cohen's d for a difference between two means: roughly 0.2 small, 0.5 medium, 0.8 large — as rough guides only
- Correlation r, or r-squared, for the strength of association
- Odds ratio, risk ratio and risk difference for binary outcomes
- Eta-squared or partial eta-squared for ANOVA
- Always interpret the effect in the units of the problem: "8 mmHg lower" means more to a clinician than " $d = 0.42$ "
- Statistical significance with a trivial effect size is a large-sample artefact, not a finding

WHAT EACH NUMBER TELLS YOU

Quantity	Answers	Depends on sample size?
p-value	Is this compatible with no effect?	Yes, strongly
Effect size	How big is the effect?	No
Confidence interval	How precisely do we know it?	Yes, width shrinks with n
Power	Could this study have detected it?	Yes

Errors, power and multiplicity

- Type I error: concluding there is an effect when there is none. Its rate is the significance level, conventionally 5%
- Type II error: missing a real effect. Its rate is beta; power is 1 minus beta
- 80% power is conventional, meaning a 1 in 5 chance of missing a real effect of the size you specified
- An underpowered study that finds nothing has not shown absence of an effect
- Testing 20 hypotheses at 5% gives roughly a 64% chance of at least one false positive
- Adjust for multiplicity (Bonferroni is simple and conservative; Benjamini-Hochberg controls false discovery rate), or pre-specify one primary outcome
- Pre-specifying the primary analysis is the cleanest defence against fishing

Worked examples

Two non-significant results that mean opposite things

Two trials of the same drug both report "no significant difference" ($p = 0.31$ and $p = 0.28$). A reviewer treats them as equivalent evidence that the drug does not work.

1. Trial A: difference 0.4 mmHg, 95% CI -0.8 to 1.6. Narrow, centred near zero.
2. Trial B: difference 6.0 mmHg, 95% CI -5.5 to 17.5. Wide, spanning clinically important effects in both directions.
3. Trial A has effectively excluded a meaningful effect — its interval contains nothing that would change practice.
4. Trial B is simply uninformative; it is compatible with the drug being useless or highly effective.
5. The p-values are almost identical and tell you none of this.
6. Reporting only " $p > 0.05$ " would make the two studies indistinguishable in a review.

THE POINT

Absence of significance is not evidence of absence. The confidence interval is what separates "we showed there is no meaningful effect" from "we could not tell", and it is why intervals belong in every result.

Twenty tests, one press release

A study measures 20 outcomes, finds one significant at $p = 0.04$, and the abstract reports it as the main finding.

1. With 20 independent tests at the 5% level, the chance of at least one false positive is $1 - 0.95^{20}$, about 64%.
2. So finding one significant result among twenty is the expected outcome under no effect at all.
3. A Bonferroni-adjusted threshold would be $0.05/20 = 0.0025$; $p = 0.04$ does not come close.
4. Benjamini-Hochberg is less conservative and controls the false discovery rate instead.
5. The cleaner protection is to pre-specify one primary outcome in the protocol, before seeing data.
6. The finding is not worthless — it is hypothesis-generating, and should be reported as exploratory.

THE POINT

Testing many things and reporting the winner inverts the meaning of the p-value. Pre-specifying the primary outcome is the cheapest defence, and it has to happen before the analysis, not after.

Key terms

Sampling distribution

The distribution of an estimate across all possible samples of a given size.

Standard error

The standard deviation of the sampling distribution; the precision of an estimate.

Confidence interval

A range constructed so that a stated proportion of such intervals contain the true value over repeated sampling.

p-value

The probability of data at least as extreme as observed, assuming the null hypothesis is true.

Effect size

A measure of the magnitude of an effect, independent of sample size.

Power

The probability of detecting an effect of a specified size when it genuinely exists.

Multiplicity

The inflation of false positive risk that comes from testing many hypotheses.

Activity

Find a paper in your field that reports "no significant difference". Locate the confidence interval. Decide whether the study showed there was no meaningful effect, or simply could not tell — and write one sentence explaining how the interval settles it.

Further reading

- **Greenland et al. (2016)** — "Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations" — Open access. Twenty-five common misreadings, each corrected. Essential.
- **Wasserstein & Lazar (2016)** — **ASA Statement on p-Values** — The profession's own statement of what p-values cannot do.
- **Sullivan & Feinn (2012)** — "Using Effect Size — or Why the P Value Is Not Enough" — Short and practical.

Self-check questions

1. A 95% confidence interval for a mean difference is 2.1 to 8.7 mmHg. Which statement is correct?
 - a. There is a 95% probability the true difference lies between 2.1 and 8.7
 - b. If the study were repeated many times, 95% of such intervals would contain the true difference
 - c. 95% of participants had a difference in that range
 - d. The result is not statistically significant
2. $p = 0.03$ means:
 - a. There is a 3% probability the null hypothesis is true
 - b. If the null were true, data at least this extreme would occur 3% of the time
 - c. There is a 97% probability the finding is real
 - d. The effect is small
3. A study of 50,000 people finds a mean difference of 0.4 mmHg with $p < 0.001$. The most sensible reading is:
 - a. A clinically important effect has been demonstrated
 - b. The effect is statistically detectable but far too small to matter clinically
 - c. The analysis must be wrong
 - d. The sample was too small

4. Testing 20 independent hypotheses at the 5% level gives roughly what chance of at least one false positive?
- 5%
 - 20%
 - 64%
 - 95%
5. A trial with 30% power finds no significant difference. You should conclude:
- The treatment does not work
 - The study was too small to detect an effect of the size expected, so it is uninformative
 - The null hypothesis is proven
 - The p-value must be wrong
6. Reporting "p = 0.000" is poor practice because:
- p-values cannot be reported to three decimals
 - A p-value is never exactly zero; it should be written $p < 0.001$
 - It implies the sample was too large
 - SPSS never produces that output
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Answers

- b.** The confidence statement is about the procedure across repeated sampling, not about this one interval. The true value is fixed; the interval is what varies. Option C confuses a confidence interval with a reference range.
 - b.** The p-value is computed assuming the null is true, so it cannot be the probability that the null is true. It is a statement about the data given a hypothesis, not about a hypothesis given the data.
 - b.** Large samples make trivial differences statistically significant. 0.4 mmHg is well inside measurement error and would change no decision. This is exactly why effect sizes must be reported alongside p-values.
 - c.** $1 - 0.95^{20}$ is about 0.64. Running many tests and reporting the significant ones is why pre-specifying a primary outcome, or adjusting for multiplicity, matters.
 - b.** With 30% power the study would miss a real effect 70% of the time. Absence of evidence here is not evidence of absence; the confidence interval will be wide and consistent with an important effect.
 - b.** Software rounds a very small value to 0.000. The probability is small, not zero, so the correct notation is $p < 0.001$.
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